

CIDR & Subnet Mask Reference Table

CIDR	Subnet Mask	Total IPs	Usable Hosts	Wildcard
/32	255.255.255.255	1	0 (Host/Loopback)	0.0.0.0
/31	255.255.255.254	2	2 (P2P Link)	0.0.0.1
/30	255.255.255.252	4	2	0.0.0.3
/29	255.255.255.248	8	6	0.0.0.7
/28	255.255.255.240	16	14	0.0.0.15
/27	255.255.255.224	32	30	0.0.0.31
/26	255.255.255.192	64	62	0.0.0.63
/25	255.255.255.128	128	126	0.0.0.127
/24	255.255.255.0	256	254	0.0.0.255
/23	255.255.254.0	512	510	0.0.1.255
/22	255.255.252.0	1,024	1,022	0.0.3.255
/21	255.255.248.0	2,048	2,046	0.0.7.255
/20	255.255.240.0	4,096	4,094	0.0.15.255
/19	255.255.224.0	8,192	8,190	0.0.31.255
/18	255.255.192.0	16,384	16,382	0.0.63.255
/17	255.255.128.0	32,768	32,766	0.0.127.255
/16	255.255.0.0	65,536	65,534	0.0.255.255
/15	255.254.0.0	131,072	131,070	0.1.255.255
/14	255.252.0.0	262,144	262,142	0.3.255.255
/13	255.248.0.0	524,288	524,286	0.7.255.255
/12	255.240.0.0	1,048,576	1,048,574	0.15.255.255
/11	255.224.0.0	2,097,152	2,097,150	0.31.255.255
/10	255.192.0.0	4,194,304	4,194,302	0.63.255.255
/9	255.128.0.0	8,388,608	8,388,606	0.127.255.255
/8	255.0.0.0	16,777,216	16,777,214	0.255.255.255

Binary Position Values (8-bit Octet)

2 ⁷ 128	2 ⁶ 64	2 ⁵ 32	2 ⁴ 16	2 ³ 8	2 ² 4	2 ¹ 2	2 ⁰ 1
Mask Shortcut: Accumulate from left to right (e.g., 128+64 = 192).							

Core Subnetting Formulas

TOTAL NUMBER OF SUBNETS 2 ⁿ (n = borrowed bits)
TOTAL HOSTS PER SUBNET 2 ^h (h = host bits remaining)
USABLE HOSTS PER SUBNET 2 ^h - 2 (Excludes Net ID / Broadcast)
BLOCK SIZE / INCREMENT 256 - Mask Value (In interesting octet)

Legacy Classful Ranges

Class	First Octet Range	Default Mask
Class A	1.0.0.0 - 126.255.255.255	/8 (255.0.0.0)
Class B	128.0.0.0 - 191.255.255.255	/16 (255.255.0.0)
Class C	192.0.0.0 - 223.255.255.255	/24 (255.255.255.0)
Class D (Mcast)	224.0.0.0 - 239.255.255.255	N/A
Class E (Exp)	240.0.0.0 - 255.255.255.254	N/A
* 127.0.0.0/8 reserved for loopback testing.		

RFC 1918 Private Spaces

IP Range Block	Default CIDR	Total IPs
10.0.0.0 - 10.255.255.255	10.0.0.0/8	16,777,216
172.16.0.0 - 172.31.255.255	172.16.0.0/12	1,048,576
192.168.0.0 - 192.168.255.255	192.168.0.0/16	65,536

4-Step Subnetting Checklist

- Find the **Interesting Octet** where the mask changes.
- Calculate **Block Size** (256 - Mask Value).
- Find the closest multiple below/equal to IP for **Network ID**.
- Add (Block Size - 1) to find the **Broadcast ID**.